

QA Strategy Document

Project Overview

This project is a Selenium-based automation framework developed for testing the OrangeHRM web application. The framework uses Java, Selenium WebDriver, TestNG, Maven, and the Page Object Model (POM) design pattern to create maintainable and reusable automated UI tests.

The objective is to validate critical HR workflows, reduce manual testing effort, and improve software quality through automated regression testing.

Testing Objectives

- Verify critical HR functionalities.
 - Detect regressions early.
 - Improve test repeatability.
 - Support continuous integration.
 - Reduce manual testing effort.
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Risk Analysis

Module	Risk	Priority
Login	High	Critical
Employee Management	High	Critical
Dashboard	Medium	High
Leave Management	Medium	High
User Session	Medium	High
Reports	Low	Medium

Testing Strategy

Functional Testing

Automated scenarios include:

- Valid Login

- Invalid Login
 - Logout
 - Dashboard Verification
 - Employee Management (if implemented)
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Regression Testing

Regression tests are executed after code changes to ensure existing functionality continues to work correctly.

Smoke Testing

Smoke tests verify that the application launches successfully and core functionality such as login is operational.

Test Pyramid

- **UI Tests (Current Project):** Login, Logout, Dashboard, Employee Management
 - **API Tests (Future Enhancement):** Authentication, Employee APIs
 - **Unit Tests (Future Enhancement):** Utility classes and helper methods
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CI/CD Strategy

Each code push should trigger:

1. Checkout repository
 2. Install Java
 3. Install dependencies
 4. Execute TestNG tests
 5. Generate reports
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Manual Testing

The following remain suitable for manual testing:

- UI appearance
 - Responsive layout
 - Accessibility
 - Cross-browser exploratory testing
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Conclusion

The framework provides reliable UI automation for critical OrangeHRM workflows while following best practices such as the Page Object Model and TestNG. Future enhancements include API automation, utility unit testing, parallel execution, and advanced reporting.